

# Integrable Systems — Exercise Sheet 15

The algebraic Bethe ansatz

Variant: Questions only

Conventions, as in Lecture 15: the Heisenberg chain of  $N \geq 2$  sites,  $L_{an}(\lambda) = \lambda + iP_{an}$ ,  $T_a(\lambda) = L_{aN}(\lambda) \cdots L_{a1}(\lambda) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}_a$ ,  $t = A + D$ ;  $|0\rangle$  is all spins up and  $|n\rangle$  has one down spin, at site  $n$ ;  $U(v_1 \otimes \cdots \otimes v_N) = v_N \otimes v_1 \otimes \cdots \otimes v_{N-1}$ ,  $t(0) = i^N U$ , and  $H = \frac{j}{2} \sum_n (P_{n,n+1} - 1) = \frac{j}{2} (i \frac{d}{d\lambda} \ln t|_0 - N)$ . Lecture 15 proves  $A(\lambda)|0\rangle = (\lambda + i)^N |0\rangle$ ,  $D(\lambda)|0\rangle = \lambda^N |0\rangle$ ,  $C(\lambda)|0\rangle = 0$ ,  $B(v)|0\rangle = i \sum_n v^{n-1} (v + i)^{N-n} |n\rangle$ ; with  $f(u) = (u - i)/u$  and  $g(u) = i/u$ , the exchange relations  $B(\lambda)B(\mu) = B(\mu)B(\lambda)$ ,  $A(\lambda)B(\mu) = f(\lambda - \mu)B(\mu)A(\lambda) + g(\lambda - \mu)B(\lambda)A(\mu)$  and  $D(\lambda)B(\mu) = f(\mu - \lambda)B(\mu)D(\lambda) - g(\lambda - \mu)B(\lambda)D(\mu)$  ( $\lambda \neq \mu$ ); and, with  $T_{11} = A$ ,  $T_{12} = B$ ,  $T_{21} = C$ ,  $T_{22} = D$ , the  $RTT$  relation entry by entry,

$$(\lambda - \mu) T_{jl}(\mu) T_{ik}(\lambda) - i T_{il}(\mu) T_{jk}(\lambda) = (\lambda - \mu) T_{ik}(\lambda) T_{jl}(\mu) - i T_{il}(\lambda) T_{jk}(\mu).$$

The momentum of an argument  $v$  is  $e^{ik} = v/(v + i)$ , and its rapidity  $-v - \frac{i}{2}$ .

**Exercise 1 (one magnon).** (i) For  $v \neq 0, -i$  show that  $B(v)|0\rangle$  is a plane wave, with amplitudes in the ratio  $e^{ik} = v/(v + i)$  from site to site, and that  $v = ie^{ik}/(1 - e^{ik})$  when  $e^{ik} \neq 1$ . (ii) Find  $B(0)|0\rangle$  and  $B(-i)|0\rangle$ , and show that  $B(v)|0\rangle / v^{N-1} \rightarrow iS^-|0\rangle$  as  $v \rightarrow \infty$ , where  $S^- = \sum_n S_n^-$ , so  $S^-|0\rangle = \sum_n |n\rangle$ . (iii) Using the exchange relations, show that for  $\lambda \neq v$

$$t(\lambda)B(v)|0\rangle = \Lambda(\lambda)B(v)|0\rangle + \frac{i}{\lambda - v} [(v + i)^N - v^N] B(\lambda)|0\rangle,$$

$$\Lambda(\lambda) = (\lambda + i)^N f(\lambda - v) + \lambda^N f(v - \lambda),$$

and deduce that for  $v \neq 0, -i$ ,  $B(v)|0\rangle$  is an eigenvector of every  $t(\lambda)$  if and only if  $e^{ikN} = 1$ . (iv) For such  $v$ , find the energy from  $\Lambda$ .

**Exercise 2 (three sites by the algebraic Bethe ansatz).** Take  $N = 3$ . (i) What are the eigenvalue of  $t(\lambda)$  and the energy on  $|0\rangle$ ? (ii) Find the regular solutions  $v$  of the Bethe equation for  $M = 1$ , their momenta  $k$  and rapidities, and show that  $\Lambda(\lambda) = 2\lambda^3 + 3i\lambda^2 - ie^{-ik}$  on each. What are the energy and the eigenvalue of  $U$ ? (iii) Letting  $v \rightarrow \infty$  in Exercise 1 (iii), show that  $t(\lambda)S^-|0\rangle = [(\lambda + i)^N + \lambda^N]S^-|0\rangle$  for every  $N$ .

**Exercise 3 (C past B, and a pairing).** (i) From the  $RTT$  relation entry by entry, with  $(i, j, k, l) = (2, 1, 1, 2)$  and  $(2, 2, 1, 1)$ , show that for  $\lambda \neq \mu$

$$C(\lambda)B(\mu) - B(\mu)C(\lambda) = g(\lambda - \mu)[D(\lambda)A(\mu) - D(\mu)A(\lambda)],$$

and that  $C(\lambda)C(\mu) = C(\mu)C(\lambda)$  for all  $\lambda, \mu$ . (ii) Let  $\langle 0|$  be the row vector of  $|0\rangle$ . Show that for  $u \neq v$

$$\langle 0| C(u)B(v)|0\rangle = \frac{i}{u - v} [u^N (v + i)^N - v^N (u + i)^N].$$

(iii) Find the limit  $u \rightarrow v$ , and show that for a real rapidity it is  $-\|B(v)|0\rangle\|^2$ : the sign  $(-1)^M$  of Lecture 15's reading on norms, at  $M = 1$ .

**Exercise 4 (the singular pair).** (i) Show that  $\{v_1, v_2\} = \{0, -i\}$  satisfies Lecture 15's Bethe equations for  $M = 2$ , for every  $N$ . (ii) Show that  $B(0) = i^N U S_N^-$ , where  $S_N^-$  lowers the spin of site  $N$ . (iii) Deduce that  $B(0)B(-i)|0\rangle = 0$  for every  $N$ , although neither factor alone annihilates  $|0\rangle$ . What does the algebraic Bethe ansatz give for this pair, and why is Lecture 15's energy formula stated for regular roots?

**Exercise 5 (two creation operators).** (i) Show that  $f(a)g(b) + g(a)g(b-a) = g(b)f(a-b)$  for  $a, b, a-b$  non-zero, and that the same identity holds for  $F(u) = f(-u)$  and  $G(u) = -g(u)$  in place of  $f$  and  $g$ . (ii) For pairwise distinct  $\lambda, v_1, v_2$ , derive from the exchange relations

$$A(\lambda)B(v_1)B(v_2) = f(\lambda - v_1)f(\lambda - v_2)B(v_1)B(v_2)A(\lambda) + g(\lambda - v_1)f(v_1 - v_2)B(\lambda)B(v_2)A(v_1) \\ + g(\lambda - v_2)f(v_2 - v_1)B(\lambda)B(v_1)A(v_2),$$

and the same for  $D$  with  $f(u) \rightarrow f(-u)$ ,  $g(u) \rightarrow -g(u)$ : the case  $M = 2$  of Lecture 15's lemma. (iii) Acting on  $|0\rangle$ , find the unwanted coefficient of  $B(\lambda)B(v_2)|0\rangle$  in  $t(\lambda)B(v_1)B(v_2)|0\rangle$ .

**Exercise 6 (Baxter's T-Q relation).** Let  $Q(\lambda) = \prod_{j=1}^M (\lambda - v_j)$  with pairwise distinct  $v_j$ . (i) Show that Lecture 15's  $\Lambda(\lambda) = (\lambda + i)^N \prod_j \frac{\lambda - v_j - i}{\lambda - v_j} + \lambda^N \prod_j \frac{\lambda - v_j + i}{\lambda - v_j}$  satisfies  $\Lambda(\lambda)Q(\lambda) = (\lambda + i)^N Q(\lambda - i) + \lambda^N Q(\lambda + i)$ . (ii) For  $M = 1$ , show that  $\Lambda$  is a polynomial if and only if  $(v + i)^N = v^N$ . (iii) For  $N = 4$  the tutorial's ground state has  $Q(\lambda) = \lambda^2 + i\lambda - \frac{1}{3}$ . Without using its roots, find  $\Lambda$  by dividing the right side of the T-Q relation by  $Q$ , and read off  $K$  and  $E$ .

**Exercise 7 ★. The next charge, on magnons.** Lecture 15's reading on the tower of charges states that for  $N \geq 4$ ,  $\frac{d^2}{d\lambda^2} \ln t|_0 = N + \sum_n [P_{n,n+1}, P_{n+1,n+2}]$ . (i) Let  $v_1, \dots, v_M$  be regular roots of the Bethe equations with  $\Psi \neq 0$ , and  $N \geq 3$ . Show that on  $\Psi$  the operator  $\frac{d^2}{d\lambda^2} \ln t|_0$  has the eigenvalue  $N + \sum_j (2i \sin 2k_j - 4i \sin k_j)$ : a sum of one function of each momentum. (ii) Compute  $\sum_n [P_{n,n+1}, P_{n+1,n+2}]$  on  $|x\rangle$ , and hence on a magnon  $\sum_x e^{ikx} |x\rangle$  with  $e^{ikN} = 1$  and  $e^{ik} \neq 1$ , and check the reading's formula on it for  $N \geq 4$ . (iii) Check (i) on the four-site ground state of Exercise 6.